Gravitational field strength

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a gravitational field to predict the force on a test mass.

SUCCESS CRITERIA - I can...

1. Represent field direction with arrows.
2. Use g=F/m and F=mg consistently.
3. Distinguish source field from the mass placed in it.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 kg object is in a field of 5 N/kg. Give its mass and weight, and distinguish the two quantities.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $g = F_g/m = GM/r^2$.
Outside a spherical source of mass M; r is distance from its center. Field direction is toward the source.

Worked reasoning: A 2 kg test object feels 6 N toward a source. The local field is g=6/2=3 N/kg. A 4 kg test object at that point feels 12 N.

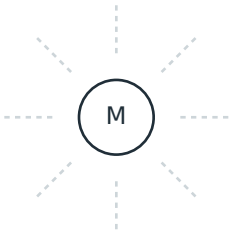
Your turn: Draw the field arrow and the two possible force arrows. Which quantity belongs to the location rather than the test mass?

Test mass (kg)	Force (N)	g (N/kg)
2	6	____
4	12	____

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw field arrows at three locations around the source. Which way would a small test mass accelerate?



Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: ① ② ③

04 Apply the physics

At one location, the gravitational field is 4 N/kg. Complete the force table for three test masses. Does changing the test mass change the field at that location?

Mass (kg)	Force (N)
1	____
2	____
3	____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student says a heavier test mass falls faster because it feels more gravitational force. Use F=mg and a=F/m to respond.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

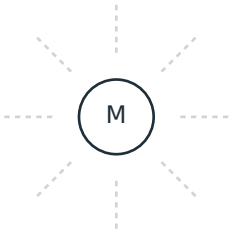
A 4 kg cart moves left at 6 m/s. Forces are 38 N right and 6 N left. Find acceleration and whether its speed initially rises or falls.
Draw the forces on the chosen object and label their sources.

Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

A 2 kg test mass feels 10 N. Find g and the force on 4 kg there.



Add arrowheads to show field direction.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two test masses experience forces in a 1: 3 ratio at the same point. Create possible masses and force values for a 5 N/kg field.

Location	Test mass (kg)	Field (N/kg)	Force (N)
1	____	____	____
2	____	____	____

Marzano check - circle: 1 2 3